

ST THOMAS CHARLOTTE AMALIE, Virgin Islands, U.S.

WMO: 994042

Lat: 18.3351N

Lon: 64.9197W

Elev: 5

StdP: 101.26

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.3	22.8	N/A	N/A	N/A	N/A	N/A	N/A	5.7	27.4	5.3	27.2	1.1	80	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.7	(c) 32.1	(d) N/A	(e) 31.7	(f) N/A	(g) 31.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.7	(p) 80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.8	4.3	DB	21.2	32.8	0.5	0.6	20.8	33.2	20.6	33.6	20.3	33.9	20.0	34.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	25.9	25.9	25.9	26.6	27.5	28.5	28.8	29.0	28.8	28.5	27.6	26.6	(6)	
(7)		DBStd	1.44	0.67	0.67	0.92	0.84	0.96	0.85	0.75	0.77	0.82	0.87	0.83	0.77	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
(10)		CDD10.0	6376	493	444	492	499	542	554	584	589	564	573	527	514	(10)	
(11)		CDD18.3	3334	234	211	234	249	284	304	326	331	314	314	277	256	(11)	
(12)		CDH23.3	35258	1809	1592	1828	2284	3055	3627	4020	4139	3888	3752	2943	2321	(12)	
(13)		CDH26.7	10628	235	199	252	414	817	1305	1587	1696	1542	1368	805	408	(13)	
(14)	Wind	WSAvg	1.7	1.7	1.8	2.0	1.9	1.7	1.8	1.9	1.7	1.5	1.5	1.5	1.8	(14)	
(15)	Precipitation	PrecAvg	996	46	36	55	65	98	74	66	100	120	140	118	78	(15)	
(16)		PrecMax	1530	94	69	144	417	286	206	143	217	326	430	284	191	(16)	
(17)		PrecMin	590	11	13	10	6	9	11	19	20	43	36	20	15	(17)	
(18)		PrecStd	275	24	16	39	83	74	56	35	54	59	103	85	41	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.3	30.0	30.3	31.2	32.0	32.2	32.6	32.5	32.2	31.2	30.1	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	28.8	28.7	29.2	29.5	30.5	31.4	31.7	32.0	31.9	31.6	30.7	29.6	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	28.5	28.4	28.7	29.1	30.1	31.0	31.3	31.6	31.4	31.1	30.3	29.1	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	28.0	27.9	28.1	28.6	29.6	30.6	31.0	31.2	31.0	30.6	29.8	28.7	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	4.7	4.5	4.3	4.2	4.4	4.6	4.7	4.6	4.6	4.8	4.6	(35)	
(36)			MADB	5.1	5.1	5.0	4.6	4.6	4.7	4.8	5.0	5.2	5.1	5.1	4.9	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.370	0.382	0.382	0.406	0.446	0.514	0.522	0.483	0.440	0.392	0.383	0.376	(40)		
(41)		taud	2.578	2.536	2.532	2.493	2.393	2.162	2.133	2.278	2.429	2.583	2.600	2.596	(41)		
(42)		Ebn at Noon	915	920	928	900	854	790	786	823	861	898	895	899	(42)		
(43)		Edh at Noon	94	103	107	112	122	152	157	137	117	97	92	91	(43)		
(44)	All-Sky Solar Radiation	RadAvg	4.77	5.57	6.21	6.56	6.33	6.36	6.45	6.35	5.96	5.22	4.59	4.42	(44)		
(45)		RadStd	0.13	0.12	0.23	0.38	0.42	0.38	0.32	0.23	0.38	0.37	0.32	0.20	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	+0.23	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+79	+80

Nomenclature: See separate page